

- Q.20 What are the advantages and applications of synchronous motor?
- Q.21 Briefly explain the working of a stepper motor.
- Q.22 Define CVT and faraday's law of electromagnetic induction.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What is a 3Phase induction motor? Write down its principle of working. How it is different from single-phase induction motor?
- Q.24 Explain the construction and working theory of a single phase synchronous motor with diagram and also write its two applications.
- Q.25 Define transformer. What is voltage regulation in transformer? Explain the losses in transformer.

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Roll No.

3rd Sem/ Instrumentation & Control Engineering Subject : Electrical Machines

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 _____ is a measure of the voltage change between no load and full load in transformers.
- a) Copper loss b) Iron loss
- c) Voltage regulation d) Torque angle
- Q.2 _____ is the most common type of feedback sensor used in servo motors.
- a) Encoder b) Potentiometer
- c) Hall sensor d) Tachometer
- Q.3 Commutator in DC generator is used for
- a) Collecting of current
- b) Reduce Losses
- c) Increase efficiency
- d) Convert AC armature current into DC

- Q.4 What are the different types of excitations in DC motors?
- Separately excited DC motor
 - Self excited DC motor
 - Permanent magnet DC motor
 - All of the above
- Q.5 Which of the following is/are the losses in transformers.
- Copper loss
 - Iron Loss
 - Dielectric loss
 - All of the above
- Q.6 _____ is a brushless DC electric motor that divides a full rotation into a number of equal, precise steps.
- Stepper motor
 - Servo motor
 - Transformer
 - None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Dielectric losses in a transformer.
- Q.8 Expand CVT.

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- Q.9 Write one application of stepper motor.
- Q.10 Define torque angle in electrical machine.
- Q.11 A single phase synchronous motor is a constant speed AC motor that operates at synchronous speed. (True/False)
- Q.12 Write one application of single phase induction motor.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What are the applications of an AC motor? Write any four.
- Q.14 Define efficiency of transformer. What is the condition for maximum efficiency in transformer.
- Q.15 Explain the process of induction of e.m.f. in a coil rotating in a magnetic field with diagram.
- Q.16 What are the factors which determines the speed of a DC motor?
- Q.17 Write down the differences between motor and generator.
- Q.18 How commercial efficiency is different from all-day efficiency of a transformer?
- Q.19 What is servo motor? How does the construction of servo motor different from stepper motor?

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